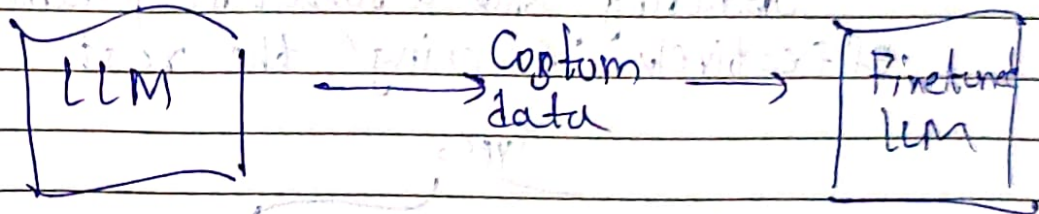


8) Fine-tuning

↳ Adapting a pre-trained model to a specific task by training the model on additional data

(Parameters, weight, batches going to change)



Fine-tuning

Instruction Fine-tuning

↳ Training a language on a set of tasks using specific instruction

'Model with instruct' as input

Classification Fine-tuning

↳ Model is trained to recognize a specific set of class labels, such as spam or no spam

No input instruction

Can handle broader set of tasks

Can handle Narrow set of data

↳ Larger Dataset, More computation Power

1) Full-fine tuning : ~~gradient-descent~~

↳ Update all model weights, creating a new version with improved capabilities.
It requires sufficient memory & computation resources, similar to pre-training.

2) Parameter Efficient Fine tuning
(More Efficient)

Updating only a subset of parameters, effectively 'freezing' the rest.

Types

LORA

QLORA

Fine-tuning

Stage 1

Dataset preparation

Stage 2

Model setup

Stage 3

Model fine-tuning & usage

1) Download the dataset



2) Preprocess dataset



3) Create data loader

4) Initialize model

5) Load pre-trained weights

6) Modify Model for fine-tuning

7) Implement evaluation utilities

8) Fine-tune model

9) Evaluate Fine-tuned model

10) Use model on new data

Classification Finetuning (Email spam)

1) Download dataset

↳ Downloading spam & ham messages from that one website

2) Preprocess

↳ Sampling balancing the data,

dividing into ~~the~~ training data (70%), validation data (20%),
testing data (10%)

3) Data loader

Dataset:

Procedure 1: Input text

↳ Tokenize

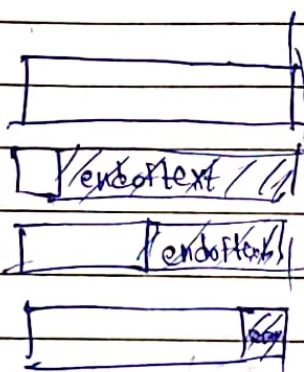
↳ Ensure that all

Sentences are same /
length

for that, make/extend

Short message equal in
length to the longest message,

Fill the short message,
with `<endoftext>` : 50256 tokens



max length

Sometimes we also need
to truncate very long
Message; till
max. length

Procedure 2: (target pair)

Create label array,

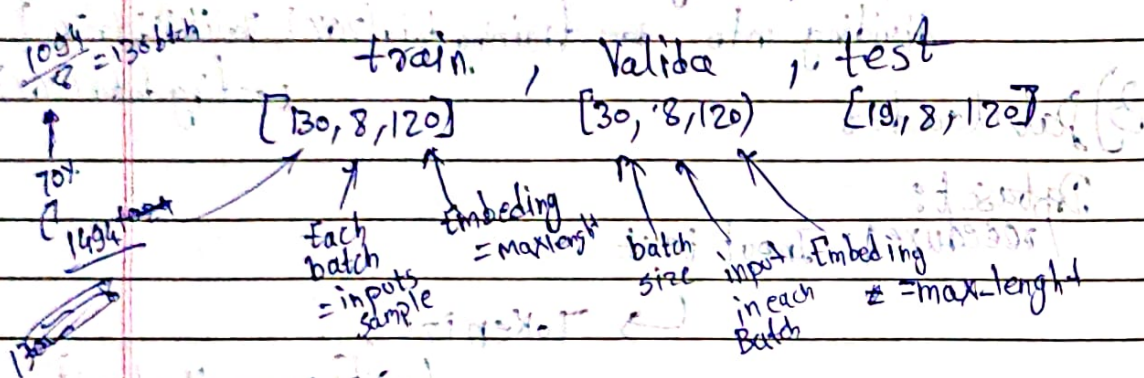
Where: 1 represent spam

& 0 represent ham

Dataset Done

It will be input for DataLoader

Create DataLoader for



1) Initialize Model

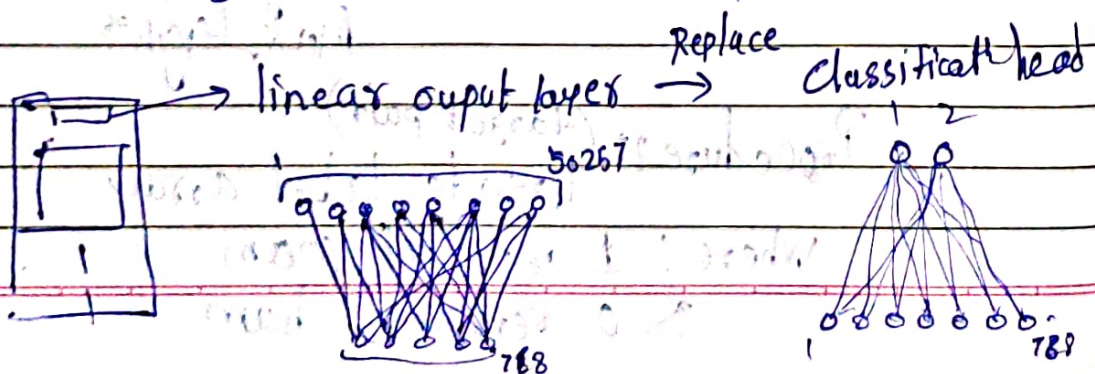
2) Load pretrained weight

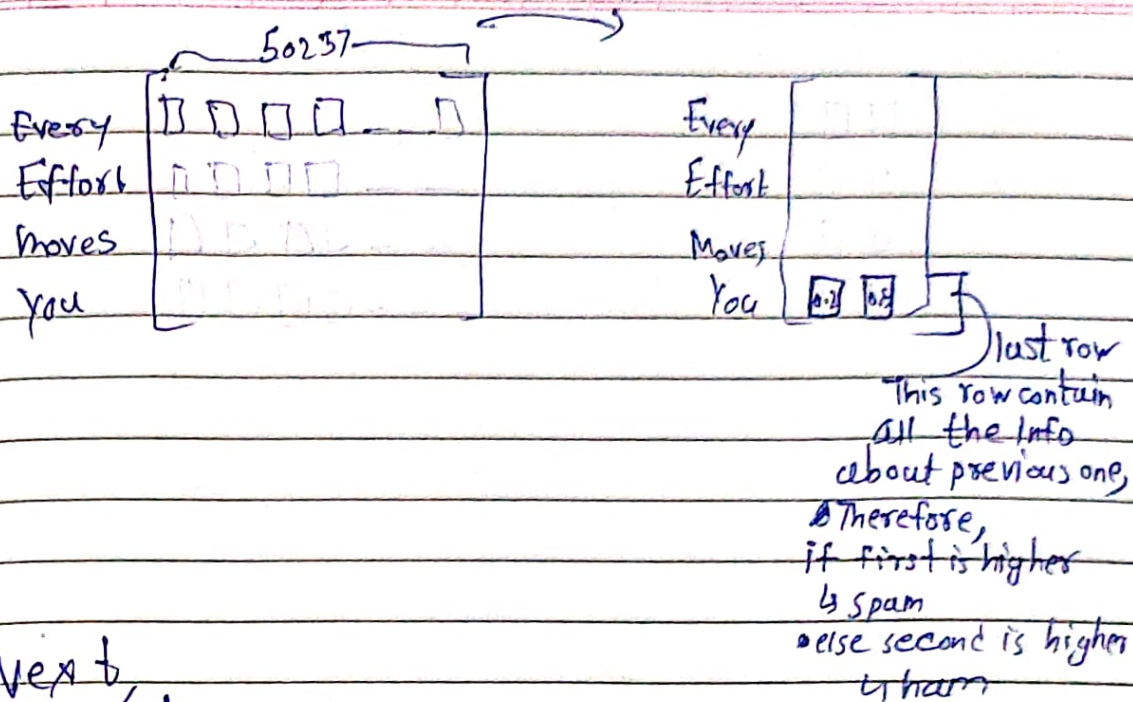
step 1: Load pretrained GPT-2 weights

Implement it.

3) Modify Model for Finetuning

Modify the architecture by adding a classification head





Next,
Select

Which layers you want to finetune

↳ Since we start with a pretrained model, it is not necessary to finetune all layers

↳ This is because the lower layers capture basic language structures & semantics which are applicable across a wide range of tasks & datasets

↳ Finetuning last layer is enough.

We will Finetune

- ↳ Final output block
- ↳ Final transformer block (12^{th})
- ↳ Final layer Norm Module

we will freeze all other parameters